Abhishu Oza

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Summary

MS Computer Science student at Rutgers University (4.0 GPA) working on AI safety: debate, honesty training and chain-of-thought monitorability. Reproduced OpenAI's confessions training method at small open scale, and replicated their monitorability evals on an open-weight stack for my MS thesis. Prior work published in IEEE Access. Seeking AI alignment research positions.

Education

Rutgers University - New Brunswick — Jan 2025 to present
Master of Science, Computer Science — GPA: 4.0/4.0
Coursework: Machine Learning I & II, Artificial Intelligence, Massive Data Mining, Computational Geometry

Ahmedabad University — Oct 2020 to May 2024
BTech in Computer Science and Engineering — GPA: 3.2/4.0
Coursework: Machine Learning, Computer Vision, Probabilistic Graphical Models, Design and Analysis of Algorithms, Applied Linear Algebra, Probability and Stochastic Processes

Research Experience

LLM Confessions Training Stress Test

- Reproduced OpenAI's confessions training (arXiv:2512.08093), an AI safety method that elicits honesty by training a reward-disjoint self-report channel, using dual-span GRPO on Qwen3-8B.
- Built a 5×5 stress-test grid (checkpoint × eval-time pressure) showing confession honesty survives instruction pressure, and ran a forced-enumeration ablation across model scales (1.7B–32B) and training stages (OLMo-3) separating capability confusion from concealment.
- Published a technical writeup, documenting where the paper's results do and do not transfer to open-weight scale.

Evaluating Chain-of-Thought Monitorability on Open Weights

- Conducted open-model replication of OpenAI's CoT-monitorability intervention evals. Built the inference layer and a concurrent eval harness over the official scaffold (gpt-oss-20B agent, Gemma 4 31B monitor, self-hosted vLLM on SLURM). Graded 8,500 samples across 6 intervention datasets.
- Initial results with monitor on low-reasoning-effort reproduce the monitorability results on an open stack (cross-fitted g-mean2 on DAFT: 0.66 CoT-only vs 0.00 answer-only).

Deep Learning for Extended Time Series Forecasting

- Researched long-term time series forecasting (DLinear, TimesNet, iTransformer) for my undergraduate thesis in the MICxN lab at Ahmedabad University. Proposed a novel fusion 1DCNN-LSTM architecture for 24-hour energy load forecasting of 100 households, published in IEEE Access [1].

Publications

[1] **A. Oza**, D. K. Patel and B. J. Ranger, "Fusion ConvLSTM-Net: Using Spatiotemporal Features to Increase Residential Load Forecast Horizon," in *IEEE Access*, vol. 13, pp. 12190-12202, 2025, doi: 10.1109/ACCESS.2025.3528072.

Skills

Machine Learning: PyTorch, TensorFlow, NumPy/Pandas, Scikit-learn, HuggingFace, vLLM, LangChain, PEFT, TRL, DDP
Programming: Python, C/C++, Java, SQL, FastAPI, Flask, PostgreSQL, Kubernetes, Docker
Tools: Git, Linux Shell, Slurm, MATLAB

Experience

Machine Learning Intern, AIML.com — Jun 2026 to present

- Developed ML projects implementing modern deep learning algorithms from scratch in PyTorch - GPT, DQN, Vision Transformer, Contrastive Learning, Distributed Optimization) with comprehensive visualizations, training and benchmarking. Authored accompanying technical articles synthesizing research papers into accessible tutorials.

Student Programmer and Part Time Lecturer, Rutgers University — May 2025 to Dec 2025

- Engineered a coding assignment system to convert DAGs and YAML specifications into Jupyter notebooks for automated assignment grading. Contributed to backend infrastructure using Flask, AWS, and PostgreSQL to serve 100s of students and teachers at the Department of Computer Science.
- Worked as a Part Time Lecturer and TA for the Introduction to Data Science course under Professor James Abello.

Machine Learning Intern, Reliance Jio Platforms — May 2023 to Jul 2023

- Engineered a VGG19-based computer vision classification model for the project "Agriculture - Crop Selection and Planning", automating satellite imagery geofencing and plant anomaly detection. Developed a geospatial web interface using the Google Maps API for visual analytics over large-scale datasets.

Projects

Documentation RAG based coding tool

- Built an RAG-based coding tool using LangChain, ChromaDB, and sentence-transformers that scrapes live documentation, indexes it into a vector store, and generates contextual code via OpenAI, Anthropic, or local HuggingFace models.
- Designed a thin-client/fat-server architecture with FastAPI and Typer CLI with remote mode via httpx, Docker deployment, optional dependency groups via pyproject.toml, and a comprehensive test suite using pytest.

GRPO Math Reasoning with Reinforcement Learning

- Applied Group Relative Policy Optimization (GRPO) to Qwen2.5-1.5B-Instruct on GSM8K via TRL's GRPOTrainer. Achieved +16.7 pt correctness and +47.0 pt formatting accuracy improvement on the full 1,319-example test set.
- Separated generation and training across 8 GPUs using vLLM as a dedicated inference server with DDP training, improving training time by 5x (from 90 hours to 18 hours). Added Liger kernel for memory optimization.

Connect 4 AlphaZero

- Implemented AlphaZero from scratch in PyTorch to learn Connect 4 through self-play reinforcement learning. Designed a ResNet with policy and value heads, guided by Monte Carlo Tree Search (MCTS) with 400 simulations per move.
- Trained using 500 self-play games per iteration with batched GPU inference across 32 parallel games, a 200K replay buffer with augmentation, and a playable GUI via pygame.

Legal Domain LoRA Finetuning using PEFT

- Finetuned Qwen3-8B on 100,000 court opinions from CourtListener using Low-Rank Adaptation (LoRA) via the PEFT library. Conducted rank ablation studies for $r \in \{4, 8, 16\}$.
- Evaluated text completion perplexity on held out court opinions and reasoning performance on LegalBench.

Miscellaneous

Graduate Record Examination (GRE) - 332/340 — 2024

- Quantitative Reasoning: 168/170, Verbal Reasoning: 164/170, Analytical Writing: 4.0/6.0.

Co-Founder and Chairman – ACM Student Chapter — 2022 to 2023

- Co-founded and chaired the ACM student chapter at Ahmedabad University.
- Hosted events and ran competitions on foundational concepts in Machine Learning and Blockchain.

Silver Certificate – Technothlon — 2019

- Secured an All-India Rank of 62 in Technothlon, a school championship organized by IIT Guwahati.